1 Closure under vertex and edge self-amalgamation

For a finite graph H and $0 \le p < 1$, define

$$\Phi_H(p) := (1-p)^{v(H)} \chi_H\left(\frac{1}{1-p}\right),$$

with the continuous extension at $p = 1$.

Since χ_H is monic of degree $v(H)$, this extension has value $\Phi_H(1) = 1$. Throughout, a graphon means a symmetric, jointly measurable function $W : \Omega^2 \to [0,1]$ on an arbitrary probability space $(\Omega, \mathcal{A}, \mu)$. We write

$$p = t(K_2, W).$$

1.1 Definitions

Definition 1 (Vertex self-amalgam)**.** *Let F be a finite graph with a distinguished vertex $a \in V(F)$, and let $N \ge 1$.*

Take N disjoint labelled copies

$$F_1, \dots, F_N$$

of the rooted graph (F, a) and identify all N copies of the distinguished vertex a, making no other identifications.

The resulting graph is denoted by

$$B_N^{(1)}(F, a)$$

and is called the N-fold vertex self-amalgam of (F, a).

Definition 2 (Edge self-amalgam)**.** *Let F be a finite graph with a distinguished ordered edge*

$$(a, b), \qquad ab \in E(F),$$

and let $N \ge 1$.

Take N disjoint labelled copies of the rooted graph (F, a, b). Identify all copies of a with one another and all copies of b with one another, respecting the order of the distinguished edge, and make no other identifications.

The resulting graph is denoted by

$$B_N^{(2)}(F, a, b)$$

and is called the N-fold edge self-amalgam of (F, a, b).

The ordering of the distinguished edge is only used to specify unambiguously how the copies are glued. It is important when the two endpoints of the distinguished edge are not interchangeable by an automorphism of the rooted graph.

1.2 The graphon gluing inequalities

We first prove the analytic part directly in graphon space.

Lemma 3 (Vertex self-amalgamation inequality)**.** *Let F be rooted at a vertex a, and let $N \ge 1$. Then for every graphon W on every probability space,*

$$t(B_N^{(1)}(F, a), W) \ge t(F, W)^N.$$

Proof. For $x \in \Omega$, define the rooted homomorphism density

$$f(x) := \int_{\Omega^{V(F)\setminus\{a\}}} \prod_{uv \in E(F)} W(z_u, z_v) \prod_{v \in V(F)\setminus\{a\}} d\mu(z_v),$$

where

$$z_a = x.$$

By Fubini's theorem,

$$t(F, W) = \int_\Omega f(x)\, d\mu(x).$$

In the graph $B_N^{(1)}(F, a)$, the N copies of $F \setminus \{a\}$ are pairwise vertex-disjoint once the image of the common root is fixed. Hence their integrations factor, giving

$$t(B_N^{(1)}(F, a), W) = \int_\Omega f(x)^N\, d\mu(x).$$

Since $f \geq 0$ and $u \mapsto u^N$ is convex on $[0, \infty)$, Jensen's inequality gives

$$\begin{aligned} t(B_N^{(1)}(F, a), W) &= \int_\Omega f(x)^N\, d\mu(x) \\ &\geq \left(\int_\Omega f(x)\, d\mu(x) \right)^N \\ &= t(F, W)^N. \end{aligned}$$

All integrands are measurable and bounded between 0 and 1. Thus Tonelli–Fubini applies, including on probability spaces that are not atomless or standard Borel. □

Lemma 4 (Edge self-amalgamation inequality)**.** *Let F have a distinguished ordered edge (a, b), let $N \geq 1$, and let W be a graphon on an arbitrary probability space with*

$$p = t(K_2, W) > 0.$$

Then

$$t(B_N^{(2)}(F, a, b), W) \geq \frac{t(F, W)^N}{p^{N-1}}.$$

Proof. For $(x, y) \in \Omega^2$, define

$$g(x, y) := \int_{\Omega^{V(F)\setminus\{a,b\}}} \prod_{uv \in E(F)\setminus\{ab\}} W(z_u, z_v) \prod_{v \in V(F)\setminus\{a,b\}} d\mu(z_v),$$

where

$$z_a = x, \qquad z_b = y.$$

The distinguished edge ab has deliberately been omitted from the product defining g. Therefore

$$t(F, W) = \int_{\Omega^2} W(x, y) g(x, y)\, d\mu(x) d\mu(y). \tag{1}$$

In the N-fold edge self-amalgam the distinguished edge is a single edge of the resulting simple graph, rather than N parallel edges. Once the images x, y of the two common root vertices have been fixed, all remaining vertices in the N copies are independent. Consequently,

$$t(B_N^{(2)}(F,a,b),W) = \int_{\Omega^2} W(x,y)g(x,y)^N \, d\mu(x)d\mu(y). \tag{2}$$

Since

$$p = \int_{\Omega^2} W(x,y)\, d\mu(x)d\mu(y) > 0,$$

the formula

$$d\rho(x,y) := \frac{W(x,y)}{p}\, d(\mu\otimes\mu)(x,y)$$

defines a probability measure on Ω^2. Equations (1) and (2) become

$$t(F,W) = p\int g\, d\rho$$

and

$$t(B_N^{(2)}(F,a,b),W) = p\int g^N\, d\rho.$$

Applying Jensen's inequality to $g \geq 0$ gives

$$\begin{aligned} t(B_N^{(2)}(F,a,b),W) &= p\int g^N\, d\rho \\ &\geq p\left(\int g\, d\rho\right)^N \\ &= p\left(\frac{t(F,W)}{p}\right)^N \\ &= \frac{t(F,W)^N}{p^{N-1}}. \end{aligned}$$

As above, the finite products are measurable and bounded, so all factorizations and changes in integration order are justified by Tonelli–Fubini. □

1.3 Chromatic-polynomial identities

The preceding analytic inequalities exactly match the chromatic-polynomial target because the densities of K_1 and K_2 are completely determined by the edge density.

Lemma 5 (Chromatic polynomial of a vertex self-amalgam)**.** *Let*

$$H = B_N^{(1)}(F,a).$$

Then

$$\chi_H(x) = \frac{\chi_F(x)^N}{x^{N-1}}.$$

Consequently,

$$\Phi_H(p) = \Phi_F(p)^N.$$

Proof. Fix an integer $k \geq 1$. By symmetry under permutations of the k colours, the number of proper k-colourings of F in which the distinguished vertex a is assigned any prescribed colour is

$$\frac{\chi_F(k)}{k}.$$

To colour H, first choose the colour of the common root, and then colour the remaining vertices in each of the N copies independently. Thus

$$\begin{aligned}\chi_H(k) &= k\left(\frac{\chi_F(k)}{k}\right)^N \\ &= \frac{\chi_F(k)^N}{k^{N-1}}.\end{aligned}$$

Because F is nonempty, $\chi_F(0) = 0$, so the factor theorem shows that x divides $\chi_F(x)$. Hence the quotient below is a polynomial. The counting identity holds for every positive integer k, so equality at infinitely many values gives the polynomial identity

$$\chi_H(x) = \frac{\chi_F(x)^N}{x^{N-1}}.$$

Let

$$n = v(F), \qquad q = 1 - p.$$

Since the N copies share exactly one vertex,

$$v(H) = Nn - (N-1).$$

Therefore, for $p < 1$,

$$\begin{aligned}\Phi_H(p) &= q^{Nn-(N-1)}\frac{\chi_F(q^{-1})^N}{(q^{-1})^{N-1}} \\ &= q^{Nn}\chi_F(q^{-1})^N \\ &= \left(q^n\chi_F(q^{-1})\right)^N \\ &= \Phi_F(p)^N.\end{aligned}$$

The identity at $p = 1$ follows by continuity; both sides equal 1. □

Lemma 6 (Chromatic polynomial of an edge self-amalgam)**.** *Let*

$$H = B_N^{(2)}(F, a, b).$$

Then

$$\chi_H(x) = \frac{\chi_F(x)^N}{(x)_2^{N-1}},$$

where

$$(x)_2 = x(x-1).$$

Consequently, for $p > 0$ (including $p = 1$ by continuous extension),

$$\Phi_H(p) = \frac{\Phi_F(p)^N}{p^{N-1}}.$$

Proof. Fix an integer $k \geq 2$. Since a and b are adjacent, every proper colouring assigns them two distinct colours.

The permutation group of the k colours acts transitively on the ordered pairs of distinct colours. Hence, for any prescribed ordered pair (c_1, c_2) with $c_1 \neq c_2$, the number of proper colourings of F satisfying

$$a \mapsto c_1, \qquad b \mapsto c_2$$

is

$$\frac{\chi_F(k)}{(k)_2}, \qquad (k)_2 = k(k-1).$$

There are $(k)_2$ choices for the ordered pair of colours on the common root edge, after which the N copies may be coloured independently. Thus

$$\begin{aligned}\chi_H(k) &= (k)_2 \left(\frac{\chi_F(k)}{(k)_2}\right)^N \\ &= \frac{\chi_F(k)^N}{(k)_2^{N-1}}.\end{aligned}$$

Because ab is an edge, $\chi_F(0) = \chi_F(1) = 0$. The factor theorem therefore shows that $(x)_2 = x(x-1)$ divides $\chi_F(x)$. Thus the quotient below is a polynomial. Since the counting identity holds for every integer $k \geq 2$, equality at infinitely many values gives

$$\chi_H(x) = \frac{\chi_F(x)^N}{(x)_2^{N-1}}.$$

Let

$$n = v(F), \qquad q = 1 - p.$$

The N copies share exactly two vertices, so

$$v(H) = Nn - 2(N-1).$$

Moreover,

$$\begin{aligned}(q^{-1})_2 &= q^{-1}(q^{-1} - 1) \\ &= \frac{1}{q}\frac{1-q}{q} \\ &= \frac{p}{q^2}.\end{aligned}$$

For $0 < p < 1$, therefore

$$\begin{aligned}\Phi_H(p) &= q^{Nn-2(N-1)} \frac{\chi_F(q^{-1})^N}{(q^{-1})_2^{N-1}} \\ &= q^{Nn-2(N-1)} \chi_F(q^{-1})^N \left(\frac{q^2}{p}\right)^{N-1} \\ &= \frac{q^{Nn}\chi_F(q^{-1})^N}{p^{N-1}} \\ &= \frac{\Phi_F(p)^N}{p^{N-1}}.\end{aligned}$$

At $p = 1$, the same identity follows by continuity, and both sides are 1. No claim involving the displayed quotient is needed at $p = 0$. □

1.4 Closure theorem

Theorem 7 (Closure under vertex and edge self-amalgamation)**.** *Let F be a finite graph, let W be a graphon, and put*

$$p = t(K_2, W).$$

Assume

$$t(F, W) \geq \Phi_F(p)$$

and

$$\Phi_F(p) \geq 0.$$

Then the following hold.

1. *For every distinguished vertex $a \in V(F)$ and every $N \geq 1$,*

$$t(B_N^{(1)}(F, a), W) \geq \Phi_{B_N^{(1)}(F,a)}(p).$$

2. *If $p > 0$, then for every distinguished ordered edge (a, b) of F and every $N \geq 1$,*

$$t(B_N^{(2)}(F, a, b), W) \geq \Phi_{B_N^{(2)}(F,a,b)}(p).$$

Proof. For the vertex self-amalgam,

$$\begin{aligned} t(B_N^{(1)}(F, a), W) &\geq t(F, W)^N \\ &\geq \Phi_F(p)^N \\ &= \Phi_{B_N^{(1)}(F,a)}(p). \end{aligned}$$

The second inequality uses

$$t(F, W) \geq \Phi_F(p) \geq 0.$$

For the edge self-amalgam,

$$\begin{aligned} t(B_N^{(2)}(F, a, b), W) &\geq \frac{t(F, W)^N}{p^{N-1}} \\ &\geq \frac{\Phi_F(p)^N}{p^{N-1}} \\ &= \Phi_{B_N^{(2)}(F,a,b)}(p). \end{aligned}$$

Again the passage to the Nth power is justified by $\Phi_F(p) \geq 0$. □

Remark 8 (Why the nonnegativity assumption is needed)**.** *The condition*

$$\Phi_F(p) \geq 0$$

cannot simply be omitted from this abstract argument. From

$$t(F, W) \geq \Phi_F(p)$$

with $\Phi_F(p) < 0$, one cannot conclude

$$t(F, W)^N \geq \Phi_F(p)^N$$

when N is even.

In the applications relevant to the Goodman-type problem this causes no difficulty. For example, the target for an odd cycle is nonnegative for $p \geq 1/2$, and the pure-chordal target is nonnegative throughout the range in which the clique-ratio proof applies.

Corollary 9 (Preservation of the required chromatic threshold)**.** *Let*

$$r := \chi(F) \geq 3.$$

Suppose that for every graphon W with

$$p = t(K_2, W) \geq 1 - \frac{1}{r-1}$$

one has

$$t(F, W) \geq \Phi_F(p) \geq 0.$$

Then every vertex self-amalgam

$$B_N^{(1)}(F, a)$$

and every edge self-amalgam

$$B_N^{(2)}(F, a, b)$$

satisfies

$$t(H, W) \geq \Phi_H(p)$$

throughout the same range

$$p \geq 1 - \frac{1}{r-1}.$$

Moreover,

$$\chi(H) = r$$

for both kinds of self-amalgam.

Proof. Because $r \geq 3$, the displayed density range has $p \geq 1/2 > 0$. The density inequalities therefore follow immediately from the preceding theorem, including its edge-amalgamation part. It remains only to verify the chromatic number.

Every self-amalgam contains a copy of F, and therefore

$$\chi(H) \geq \chi(F) = r.$$

For a vertex self-amalgam, take a proper r-colouring of each copy. By permuting the r colour labels separately in each copy, we may arrange that the distinguished vertex has the same colour in every copy. These colourings then combine into a proper r-colouring of the amalgam. Thus

$$\chi(H) \leq r.$$

For an edge self-amalgam, the two endpoints of the distinguished edge receive distinct colours. The symmetric group on the r colours is transitive on ordered pairs of distinct colours. Hence, after permuting the colour labels independently in each copy, we may arrange that the ordered pair of colours on the distinguished edge is the same in every copy. The colourings then combine to give a proper r-colouring of the amalgam.

Thus in either case

$$\chi(H) = r.$$

Consequently the threshold

$$p \geq 1 - \frac{1}{\chi(H) - 1}$$

is exactly the original threshold

$$p \geq 1 - \frac{1}{\chi(F) - 1}.$$

□

1.5 The six-vertex Atlas application

Corollary 10 (Atlas 115)**.** *NetworkX Atlas graph* 115*, with graph6 code* `E‘Xg`*, is positive: for every graphon* W *of edge density* $p \in [1/2, 1]$*,*

$$t(F_{115}, W) \geq p^3(2p-1)^2 = \Phi_{F_{115}}(p).$$

Its order, size, and chromatic number are respectively 6*,* 7*, and* 3*, and*

$$\chi_{F_{115}}(x) = x(x-1)^3(x-2)^2.$$

Proof. Atlas 15 is the paw, with edge list

$$03,\ 12,\ 13,\ 23,$$

where 123 is its triangle and 0 is the leaf at 3. Its chromatic polynomial and target are

$$\chi_{F_{15}}(x) = x(x-1)^2(x-2), \qquad \Phi_{F_{15}}(p) = p^2(2p-1).$$

The rooted triangle–tree theorem (the case L_1) proves $t(F_{15}, W) \geq \Phi_{F_{15}}(p)$ for every $p \in [1/2, 1]$; the target is nonnegative on this interval.

Take two copies of the paw and identify, in order, the endpoints of the triangle edge 12, the edge opposite the leaf-bearing vertex 3. The result is $B_2^{(2)}(F_{15}, 1, 2)$. One convenient edge labelling of this amalgam is

$$01,\ 03,\ 05,\ 13,\ 15,\ 23,\ 45.$$

It is isomorphic to Atlas 115, whose NetworkX edge list is

$$01,\ 14,\ 15,\ 23,\ 24,\ 25,\ 45.$$

For example, the first list is carried to the second by

$$0 \mapsto 4, \quad 1 \mapsto 5, \quad 2 \mapsto 0, \quad 3 \mapsto 1, \quad 4 \mapsto 3, \quad 5 \mapsto 2.$$

The edge-amalgamation identities give

$$\chi_{F_{115}}(x) = \frac{\chi_{F_{15}}(x)^2}{x(x-1)} = x(x-1)^3(x-2)^2$$

and, since $p \geq 1/2 > 0$,

$$t(F_{115}, W) \geq \frac{t(F_{15}, W)^2}{p} \geq \frac{\Phi_{F_{15}}(p)^2}{p} = p^3(2p-1)^2.$$

The amalgam contains the paw and is 3-colourable by matching the ordered colours on the common edge, so its chromatic number is 3. At $p = 1/2$ the target is 0; at $p = 1$ both the target and the density are 1. This proves the complete required interval. □

The other two edge orbits of the paw give Atlas 111 and Atlas 117: gluing on a triangle edge incident with the leaf-bearing vertex gives the former, while gluing on the pendant edge gives the latter. Both were already positive by the cone-over-forest theorem. Exact enumeration of the scoped Atlas graphs with at most six vertices shows that Atlas 115 is the only formerly uncovered row added by vertex or edge self-amalgamation of the accepted smaller positive bases; the explicit isomorphism above supplies the proof-relevant identification.

Remark 11 (Why only K_1 and K_2 give an automatic closure)**.** *There is a useful reason that vertex and edge amalgamation are particularly clean.*

Suppose instead that copies of F are amalgamated along a rooted clique K_s. Provided $t(K_s, W) > 0$, the same normalized-measure argument gives

$$t(H, W) \geq \frac{t(F, W)^N}{t(K_s, W)^{N-1}}.$$

On the chromatic-polynomial side, whenever $\Phi_{K_s}(p) > 0$, one obtains

$$\Phi_H(p) = \frac{\Phi_F(p)^N}{\Phi_{K_s}(p)^{N-1}}.$$

For $s = 1$ and $s = 2$,

$$t(K_1, W) = 1 = \Phi_{K_1}(p), \qquad t(K_2, W) = p = \Phi_{K_2}(p),$$

so the analytic and chromatic denominators agree exactly.

For $s \geq 3$, however, $t(K_s, W)$ is not determined by the edge density p. Consequently the same argument does not by itself imply the Goodman-type target. This is precisely where additional clique-density or clique-ratio information is required.

1.6 Inputs, endpoint audit, and exact limitations

The closure theorem itself uses only Tonelli–Fubini, Jensen's inequality on a probability space, and the elementary colouring counts above. Corollary 10 additionally uses exactly the accepted rooted triangle–tree bound for the paw L_1; it does not use universal-vertex lifting, a finite-step reduction, or a regularity assumption.

The division by p^{N-1} in edge amalgamation is made only when $p > 0$. This is automatic in the catalogue application and in the required threshold corollary because $\chi(F) \geq 3$. The endpoint $p = 1$ is not obtained by substituting q^{-1} with $q = 0$: the target identities there follow from their polynomial continuations, while the graphon inequality itself follows directly from Jensen. Vanishing rooted densities cause no division, since the normalized measure $d\rho = W\, d(\mu \otimes \mu)/p$ depends only on $p > 0$.

Equality in the analytic vertex-amalgamation inequality occurs precisely when the rooted density f is constant almost everywhere (apart from the tautological case $N = 1$). Equality in the edge version occurs when g is constant ρ-almost everywhere. These observations are not needed for positivity but identify the only possible slack in the closure step.

Finally, the theorem applies only to copies that meet in exactly the specified root vertex or root edge and have no other vertices or edges between different copies. It proves neither amalgamation along a larger subgraph nor an operation in which different root types are mixed. The clique discussion above explains why the same one-line comparison does not automatically extend to K_s for $s \geq 3$.

2 What the Lean formalization actually proves

The machine-checked development is `lean/Taeyoung/Methods/SelfAmalgam.lean`, ending at `satisfiesLowerBou`

Scope: the single instance, not the closure theorem. Corollary 10 is proved in full, on all of $p \in [1/2, 1]$. The general statements are *not* formalized: neither the vertex self-amalgamation inequality, nor the edge one at general N, nor the two chromatic-polynomial identities, nor the closure theorem, nor the threshold-preservation corollary. What is formalized is the $N = 2$ edge amalgam of the paw, which is the only member of the family inside the six-vertex catalogue.

The rooted factor was already in the project. The function $g(x, y)$ of the edge self-amalgamation proof — the paw density with its distinguished edge deleted — is, for the paw,

$$g(x, y) = \int_\Omega W(x, z)W(y, z)d(z)\, d\mu(z),$$

and that is exactly `Link.pageOp W 1`, built earlier for the page-rooted book-leaf family. So both identities the proof needs are existing lemmas: (1) is `Link.integral_edge_pageOp` at $s = 1$ (`SelfAmalgam.integral_edge_page_eq`), and (2) is one six-coordinate peeling (`SelfAmalgam.homDensity_amalgam`). Nothing about amalgamation as such is formalized; the two densities are computed directly.

The departure: no measure ρ, and no hypothesis $p > 0$ on the analytic step. At $N = 2$ Jensen against the probability measure $d\rho = W\, d\mu^2/p$ *is* Cauchy–Schwarz, so the formalization never constructs ρ. It applies the project's own `PureChordal.integral_mul_sq_le_integral_mul_integral_mul_s` to the *unnormalised* weight W on $\mu \otimes \mu$, giving

$$\left(\int_{\Omega^2} Wg\right)^2 \le \left(\int_{\Omega^2} W\right)\left(\int_{\Omega^2} Wg^2\right) = p\, t(A_{115}, W) \qquad (\texttt{SelfAmalgam.sq_edge_page_le}),$$

with no hypothesis on p at all. The division by p happens once, at the very end, where $p \ge 1/2 > 0$ is already available. Consequently the note's standing assumption $p = t(K_2, W) > 0$ in the edge self-amalgamation lemma is not needed in the form used here.

The chromatic polynomial does not come from the amalgamation identity. The note derives $\chi_{F_{115}} = \chi^2_{F_{15}}/\big(x(x-1)\big)$ from the general identity for $\chi_{B_N^{(2)}}$. The formalization instead builds $\chi_{F_{115}}(x) = x(x-1)^3(x-2)^2$ from an `attachVertex` tower, the way every other catalogue row builds its chromatic polynomial: K_3 on $\{0, 1, 2\}$, then the leaf 3 on $\{2\}$, then the second apex 4 on the glued edge $\{0, 1\}$, then the leaf 5 on $\{4\}$. The root list $[0, 1, 1, 1, 2, 2]$ then gives $\Phi = p^3(2p - 1)^2$ through `chromaticTarget_affineProd`, with no per-row polynomial computation — and it is the same root list as Atlas 119.

The paw input. $t(F_{15}, W) \ge p^2(2p - 1)$ is `RootedTriangleTree.rootedTree_bound` at $r = 1$, applied to `Chromatic.pawGraph` through `paw_factorization`. That lemma is stated for Ω : Type rather than Type*, which is why this file fixes Ω : Type throughout; `SatisfiesLowerBound` quantifies over Type as well, so nothing is lost.